Interpreting questions with epistemic biases and tracking questioner beliefs in online processing

The present study investigates the processing of negated polar questions in English and German. Using a self-paced reading paradigm, it tests the hypothesis that the morphosyntactic form of such questions signals epistemic biases which affect incremental interpretation. However, we find very limited evidence in support of the hypothesis, with results in both English and German suggestive that interpreting beliefs and biases from biased questions in real time requires strong contextual and communicative pressures from which comprehenders can infer pragmatic meaning. We discuss our results in light of related psycholinguistic work on the processing of information structural cues as well as theoretical work on the interpretation of questions with biases.

Keywords: negated polar questions, biased questions, beliefs and epistemic biases, question processing, incremental interpretation, self-paced reading

1. Introduction

Previous psycholinguistic work has shown that online language processing can be guided by pragmatic considerations to do with the use of non-canonical utterance forms in

context, where both syntactic (e.g. Kaiser & Trueswell, 2004; Yano & Koizumi, 2018) and prosodic (e.g. Kurumada, Brown, Bibyk, Pontillo, & Tanenhaus, 2014; Roettger & Franke, 2019) modulations of the linguistic signal can serve as information structural cues to meaning. Crucially, in both cases the relevant cues to interpretation are derived not primarily from the usage of particular lexical items but rather from modulations in the form of the utterance which ultimately convey pragmatically relevant information. In other words, interpretation in such cases is assumed to be guided by more or less fine-grained sensitivities to do with what particular utterance formulations might signal in context. Most research in this tradition has focused on the resolution of reference in declarative sentences (e.g. *The bird ate the worm*), whose prototypical discourse function is to assert content. Yet assertions only make up one portion of the discursive landscape of language use: another essential aspect of interpretation relates to how language users make sense of questions.

In this paper, we present a study designed to investigate this further. Specifically, we look at *biased questions* in English and German, where the semantic and pragmatic literature has identified that different syntactic and prosodic realisations can carry different sorts of (normative) biases (e.g. Farkas & Roelofsen, 2017; Krifka, 2017; Van Rooy & Safarova, 2003). For example, in (1), the different morphosyntactic forms of negation are said to index different questioner beliefs in both English and German (for details, see section 2).

(1) a. Is**n't** there a vegetarian restaurant around here? (Ladd 1981)

b. Is there **not** a vegetarian restaurant around here?

c. Gibt es **nicht ein** vegetarisches Restaurant in dieser Ecke?
gives EXPL not a vegetarian restaurant in this corner
"Isn't there a vegetarian restaurant around here?"

d. Gibt es **kein** vegetarisches Restaurant in dieser Ecke?
gives EXPL not a vegetarian restaurant in this corner

"Is there not a vegetarian restaurant around here?"

(Büring & Gunlogson 2000)

Using a self-paced reading paradigm, we test the hypothesis that these morphosyntactic forms signal epistemic biases that can affect incremental interpretation. However, we find very limited evidence in support of the hypothesis, contradicting results of production studies in which there are strong preferences for speakers to produce particular question forms in contexts in which prior beliefs are made salient. Both our English and German results suggest that rather than biases being strongly related to the morphosyntactic structure, interpreting beliefs and biases from biased questions in real time requires strong contextual and communicative pressures from which comprehenders can infer pragmatic meaning.

The paper is laid out as follows. In section 2, we give a brief overview on existing literature on the semantics and pragmatics of question bias, and outline the form-meaning mappings that have been proposed—both in the theoretical and experimental literature. We then outline the relevant questions for the current study, and our hypotheses, as well as limitations regarding the design of the materials. In section 3 we present the design and results for our English study, and the German study is then presented in section 4. Finally, in section 5 we discuss the implications of the results for our understanding of question interpretation, and of utterance processing more generally, before concluding in section 6.

2. Question forms as cues to interpretation

Questions have received a considerable deal of attention in research concerned with meaning in language (e.g. Groenendijk & Stokhof, 1984; Karttunen, 1977; Raymond, 2003; Searle, 1969; Stenström, 1984). Questions are canonically associated with the interrog-

ative clause type—demonstrated in English through subject-verb inversion (2)—which often also has particular prosodic contours associated with it. The English example in (2) would be expected to exhibit rising intonation (e.g. Bartels, 1999; Hedberg, Sosa, & Görgülü, 2017).

(2) Did you buy the present?

An example like (2) is semantically taken to denote a set of propositions (Hamblin, 1973; Karttunen, 1977), and in answering the question, the addressee might be said to signal which proposition from the set they believe holds in the world.

However, questioning is not always strictly tied to a canonical morphosyntactic or prosodic structure, nor to a straightforward semantic meaning. A growing body of research is now focused on so-called non-canonical questions (Dayal, 2016), which may differ from canonical ones both in terms of their form and function. From a processing perspective, non-canonical questions are therefore interesting exactly as they do not behave according to the standard definitions of what a question is. For example, questions can also appear in declarative form – such as in indirect requests for information like (3). Questions can also serve functions which are not directly or primarily tied to information-seeking—such as in rhetorical questions like (4), which have often been claimed to be pragmatically equivalent to assertions of the opposite polarity (e.g. Han, 2002; Sadock, 1974)[1].

(3) I was wondering if you bought the present.

(4) Didn't I tell you it would be easy?!

Importantly for this paper, questions can also be non-canonical in the sense that, even

[1]More recent accounts of rhetorical questions claim that there is an assertive flavour to the interrogative which arises through presupposition, rather than a direct semantic equivalence with assertions (Biezma & Rawlins, 2017)

though they do serve as requests for information, they are biased with regard to what the relevant answer(s) might be. Questions with biases have been well-studied in semantics and pragmatics, both from a formal and a functional perspective (see De Ruiter (2012) as well as Romero (2020) for overview and discussion). Research in the formal semantic tradition has suggested that questions can carry different sorts of (normative) biases, which might, in turn, be mapped onto different syntactic and prosodic realizations.

Negated polar questions in English like (1-a), for instance, have been argued to request confirmation of something the questioner believes to be true (i.e. that there is a restaurant around the area), or express uncertainty regarding the same belief in light of evidence that said belief is actually false (i.e. that there is, in fact, no restaurant around the area). More specifically, the usage of negated polar questions has been said to be constrained both by an *original speaker bias* (Ladd, 1981; Romero & Han, 2004), which relates to whether the questioner has any (prior) belief or expectation that the proposition targeted by their question is true, as well as by *contextual evidence*, which relates to the types of evidence possibly available in the discourse situation[2].

Semantically, the difference in interpretation is often taken to be due to a difference in scope of negation. In the instance where the questioner has an original bias, and there is no challenging contextual evidence, the negation scopes *outside* the proposition (*outer negation*) and the speaker's belief in the positive proposition is put up for question or confirmation. When the speaker's original bias is challenged by some negative contextual evidence, the negation is *inside* the proposition (*inner negation*), and it is this negative proposition that is questioned (Ladd, 1981). While Ladd's original claim about interpreting negation in polar questions concerned questions in English only, similar claims about a relationship between negation and epistemic bias have been made for range of languages, including German (Büring & Gunlogson, 2000), Hungarian (Gyuris,

[2]See Romero and Han (2004) and the references therein for the original definitions of speaker bias and contextual evidence.

2016), Bulgarian, Spanish, Greek, Korean (Romero & Han, 2004) and Japanese (Sudo, 2013).

Crucially for the purposes of the study at hand, the morphosyntactic form of the question is said to index the different meanings. For example, in English, when the negation marker is realized as *not*, separately from the auxiliary verb, as in (1-b), it is termed *low negation*, and the question is said to index no particular bias from the side of the questioner (Romero & Han, 2004). It would therefore only be used with the semantic inner negation, where the negative proposition is being questioned due to contextual evidence favouring the a negative proposition. When the negation marker is realized as the contracted form *-n't*, as in (1-a), it is known as *high negation*, and is said to index a positive questioner bias towards the truth of the underlying proposition, with a semantic ambiguity between the outer and inner negation interpretations depending on the contextual evidence surrounding the interaction[3].

However, despite these claims of strong morphosyntactic indexing in the theoretical literature, empirical tests on speaker's choice of negative form in production have varied. Investigating the roles of speaker bias and contextual evidence in the production of question forms, Roelofsen, Venhuizen, and Weidman Sassoon (2012) found no difference in when speakers chose to produce high or low negation questions in English. Both were equally preferred in situations where the speaker had a prior belief and there was negative contextual evidence challenging this belief, or no particular contextual evidence either way, challenging the claim that only high negation questions encode an original speaker bias[4].

[3]Though see Sailor (2013) for experimental data and evidence from NPIs in which he argues there is no ambiguity for speakers in American English: only the outer negation interpretation is available for high negation. This argument is also pursued by Thoms, Adger, Heycock, Jamieson, and Smith (2023) for Scottish English.

[4]Note that there were some issues with the item design in Roelofsen et al.'s (2012) study, namely that in some cases the contextual evidence was not strong enough, which may have impacted the results. See Domaneschi, Romero, and Braun (2017) and Jamieson and Macuch-Silva (to appear) for discussion.

However, more recently, the two factors have been found to interact to give rise to different usage profiles by Domaneschi et al. (2017), who investigated question production in the laboratory in both English and German. Domaneschi and colleagues' motivated their comparison of English and German as "though the syntax of the English and German polar question forms is largely parallel, their pragmatic use is partly misaligned" (Domaneschi et al., 2017, p. 10). That is, while in English the high negation form *-n't* is said to be ambiguous between the inner and outer negation readings, depending on the contextual evidence, in German, the high negation marker *nicht* (as in (1-c)) is said to only license the outer negation reading, unambiguously signalling a neutral context with no evidence against the speaker's original bias.

For example, in a situation where someone is at a ticket office trying to buy a train ticket for the next day, and they believe that there is a train in the early morning, an utterance like (5) constitutes neutral evidence with regard to person's original belief, whereas (6) is evidence against that belief.

(5) "Do you have any preference?"

(6) "The only train available is at 11:00"

Now, in English, a question like *Isn't there a train in the early morning?* is said to be a felicitous discourse continuation to both (5) and (6), whereas its German equivalent *Gibt es nicht einen Zug früh am Morgen?* is said to be felicitous only as a response to a neutral antecedent like (5).

In their production study, Domaneschi and colleagues presented participants with discourse contexts which included in-built biases and contradictory contextual evidence, and asked participants to choose the most appropriate question from a list to produce, given the context. Both in English and in German, participants were more likely to produce questions with high negation—realized with *-n't* and *nicht (ein)*, respectively—

in contexts in which they were given a prior belief about the truth of the proposition, and there was contextual evidence against that very same proposition, as well as in contexts in which the evidence was neutral. As for questions with a so-called low negation—realized with *not* in English and *kein(e)* in German—participants chose to produce them in contexts in which the absence of a particular belief or expectation about the proposition was combined with negative contextual evidence against the truth of the proposition. These results can be summarized by the empirical distribution shown in Table 1:

Table 1: Attested pragmatic profile of negated polar questions, as per the data by Domaneschi et al. (2017).

		Original bias		
		p	*neutral*	$\neg p$
Contextual evidence	p			
	neutral	high negation		
	$\neg p$	high negation	low negation	

These results provide evidence to the claim that a question like *Isn't there a restaurant around here?* can be used in English to confirm a belief that something is true. By the same token, they also provide evidence to the claim that the same questions can be used to disconfirm an assumption made on the basis of evidence against a positive belief. Domaneschi et al.'s results have since been replicated and extended upon in work on Italian (Di Maro, Origlia, & Cutugno, 2021), providing further experimental evidence that signaling epistemic stance via negated questions might be a cross-linguistically common pragmatic strategy.

Although the experimental evidence suggests there may be distinctions in the choice of question form *produced* in a forced choice task given a particular set of biases and beliefs, limited evidence from processing does not mirror this result. In the visual world paradigm presented in Tian, van Tiel, Clin, and Breheny (2021), hearing questions with

English high negation *-n't* and low negation *not* led participants to consider both the p and $\neg p$ target images throughout the time course of the question, as compared to positive polar questions, in which participants settled on the p target image by the object DP. This result would seem to suggest that the form of the negation in the question did not allow participants to make predictions about the underlying proposition and the speaker's belief about it. However, the items in Tian et al. (2021) were presented out of the blue, without a prior discourse context, and so further research is required to understand the relationship between the morphosyntactic form of the question, and the biases that it indicates, in relation to the predictions made by the theoretical literature. All in all, the current empirical picture raises several interesting questions about the relationship between the morphosyntactic form of the question and the bias and beliefs that it signals.

First and foremost, from a theoretical perspective, one might ask what the implications of the attested production preferences might be for question interpretation, on the one hand, and for any underlying form representations, on the other. Can interpreters infer the relevant pragmatic biases from the forms of polar questions, or are the relevant epistemic meanings only weakly associated with them? If they are to some extent conventionally associated with particular polar questions, how do these meanings impact the processing of language and the interpretation of a speaker's commitment to the content of a question?

In the present study we address exactly the issue of whether pragmatic preferences evidenced in the production of negated polar questions affect how they are interpreted incrementally during online language comprehension. Specifically, we focus on the effect of perceived epistemic bias, or lack thereof, of the speaker, given negative contextual evidence—i.e. the bottom row of Table 1.

We hypothesise, based on the predictions from the theoretical literature, and the ex-

perimental production evidence from Domaneschi et al. (2017) and Di Maro et al. (2021), that presence or absence of an epistemic bias will affect incremental question interpretation differently depending on the syntactic form of the target question. Importantly, this entails an interaction between epistemic bias and negation type, such that high negation questions are expected to be processed more easily in discourse contexts with a particular questioner belief compared to discourse contexts without any belief, whereas low negation questions are expected to be processed more easily in contexts without any questioner belief compared to contexts with a belief.

The reported study was pre-registered under `https://osf.io/3zk6d` and, unless otherwise noted, all analytic choices and procedures follow the pre-registered analysis plan. The study was approved by the Linguistics and English Language Ethics Committee at the University of Edinburgh, reference no: XXXXXX.

3. Experiment 1 (English)

3.1 Norming task

In the main task described in section 3.2 below, we test the extent to which comprehenders draw on pragmatic expectations about what the form of a negated question encodes about the epistemic status of the questioner, and how this may impact the incremental interpretation of the question in context. Given that our aim was to systematically tease apart any potential effects of epistemic biases on online question interpretation, we normed the discourse contexts used in the processing task to make sure they indeed give rise to the epistemic expectations of interest. We briefly detail the procedure of the norming task, before moving on to the main part of the study.

3.1.1 Method

3.1.2 Participants

Forty self-reported native speakers of English were recruited using the crowdsourced platform Prolific. Participants were compensated monetarily for their participation in the experiment. Data collection was conducted entirely online and all collected data was stored at servers from the Osnabrück University.

3.1.3 Materials and design

Our key assumption in the reading task is that the modulation of the online processing signatures is contingent on the relevant questions being interpreted against contexts with particular epistemic biases, namely contexts containing a prior belief about the proposition targeted by the question or contexts containing no particular belief about the proposition targeted the question. Contexts with an overt prior questioner belief, on the one hand, should be naively interpreted as supporting whatever state of affairs is targeted by the question in consideration. For instance, in (7), a friend is adopting a new pet. The discourse context indicates that the questioner has a belief that that pet is a cat. This information is not inferred from contextual cues, but rather directly reported.

(7) Our friend is adopting a new pet. You heard from her sister that it would be a cat.

Negated question: Isn't she / Is she not adopting a cat?

A naive interpreter asked whether or not they think the person in the scenario is adopting a cat should respond with "yes", given that the context biases the positive state of affairs targeted by the question. Now compare this situation to a situation where there is no

particular overt questioner belief, like in (8) below. In comparison to (7), the person in the scenario in (8) has not received any direct information about the proposition being questioned. Contextually, the fact that the venue is "their favourite" may indicate some likelihood of having a ticket, but the concert being "sold out" may indicate the opposite. Regardless of these contextual hints, there is no direct report regarding the proposition.

(8) We are talking about a sold out concert that's coming up at our favorite venue.
Question: Didn't you / Did you not buy a ticket?

A naive interpreter asked whether or not they think the person in the scenario bought a ticket would likely respond with either "Don't know" or "No", given the relative uncertainty about the actual state of affairs.

Participants were asked to read 20 fictional scenarios, each consisting of one or two short written statements relating to a common situation such as going to a restaurant or getting a new pet—as in the examples in (7)-(8). Participants' task was to answer a question pertaining to each scenario; crucially, one which was a paraphrase of the relevant negated question investigated in the main experiment.

(9) Our friend is adopting a new pet. You heard from her sister that it would be a cat.
Question: Do you think the person in this scenario is getting a cat?

Participants saw 10 scenarios designed to contain a belief regarding the state of affairs targeted in the question (*prior belief* items) and 10 designed to contain no explicit belief of any sort (*no belief* items). The same three response options were available across all trials, namely "Yes", "No", or "Don't know". These options were counterbalanced across participants.

3.1.4 Procedure

Written instructions were provided prior to the actual task. Each trial, participants were presented with a scenario, a question pertaining to that scenario, as well as three response options, as described above. Participants were instructed to answer the question on the basis of the information available to them in the scenario. Upon clicking on one of the options and thus making a choice, participants were automatically directed to the next trial.

3.1.5 Results

The data and analysis scripts are available for inspection under `https://osf.io/7a2jk/`. Recall that the expectations were that *Prior belief* items should show a bias for "Yes" responses while *No belief* items should show a bias against "Yes" responses. We consider any ratio of "Yes" to "No"/"Don't know" responses which is higher than 40/60, depending on the expected direction of the effect, as evidence for a bias in interpretation. We set this as an arbitrary threshold, where a 50/50 split in responses would indicate no particular bias. Figure 1 shows the item norms aggregated over all items and participants[5]. Participants interpreted *Prior belief* items, on average, as reflecting specific beliefs about the given discourse scenarios. By the same token, they interpreted *No belief* items, on average, as reflecting no particular prior beliefs about the discourse scenarios in those items. All in all, these results show that the normed items do reflect the general epistemic expectations which are of theoretical relevance to the account of question interpretation entertained in this study. We thus now proceed in reporting the main processing study.

[5]Note that for each individual item, the overall result was also in the expected direction, except for one item where no bias was found.

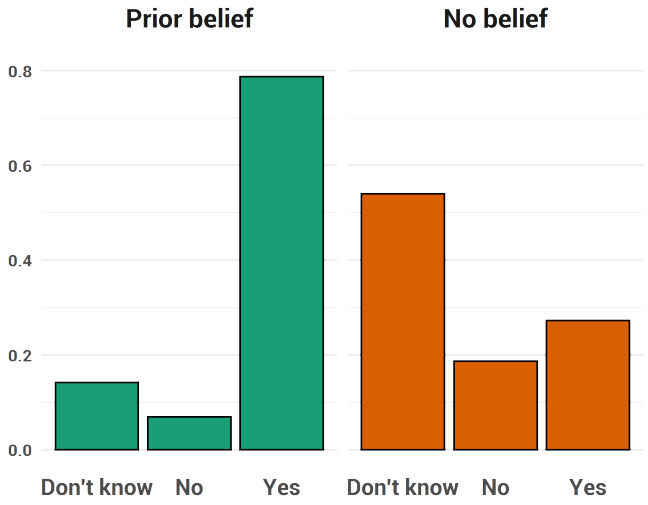

Figure 1: Item norms plotted by context type, *Prior belief* context shown in green, *No belief* context shown in orange. Each bar shows the proportion of choices of each response option, labeled individually below the respective bars.

3.2 Processing task

In the main task of the present study, our goal was to look at how English natives interpret negated questions with epistemic biases incrementally during processing. We now turn to the details and setup of this task.

3.2.1 Method

3.2.2 Participants

120 self-reported native speakers of English were recruited using the crowdsourced platform Prolific. Participants were compensated monetarily for their participation in the experiment. Data collection was conducted entirely online and all collected data was stored at servers from the Osnabrück University.

3.2.3 Materials and design

Participants were asked to read 35 fictional scenarios, each consisting of a short paragraph relating to a common situation such as going to a restaurant or getting a new pet as well as, crucially, a test sentence prefaced by another short sentence (see example item (10)). Participants' task was to read each scenario in its entirety, according to the procedure described below. There were 10 scenarios designed to contain a belief regarding the state of affairs targeted in the target sentence—which always consisted of a negated question—and 10 designed to contain no explicit belief of any sort. Every scenario also included some piece of information providing negative evidence regarding the state of affairs targeted in the target sentence.

(10) **Example item (*Prior belief, high negation*)**

Our friend is getting a new pet. You heard from her sister it would be a cat. However, I tell you she's planning to take it for a lot of walks. You say:

Hold on. Isn't she getting a cat?

In (10), *You heard from her sister it would be a cat* is the belief, *However, I tell you she's planning to take it for a lot of walks* is the negative contextual evidence, and *Isn't she getting a cat?* is the target sentence. Each target sentence conformed to the sentence frame *auxiliary* | *pronoun* | *(negation marker)* | *verb* | *verb spillover* | *noun*, and was preceded by a short utterance in order to ensure that the first region was not a region of interest. In (11) below, *We are talking about a sold out concert that's coming up at our favorite venue* is the utterance without any belief, *I say I am not going* is the negative contextual evidence, and *Didn't you buy a ticket?* is the target sentence.

(11) **Example item (*No belief, high negation*)**

We are talking about a sold out concert that's coming up at our favorite venue.

I say I am not going. You say:

That's a shame. Didn't you buy a ticket?

Any individual participant saw each scenario either with a high or a low negation question as a target, e.g., a participant would see (11) either with *Didn't you buy a ticket?* or with *Did you not buy a ticket?*. In addition to 20 the critical trials, there were 15 fillers containing non-negated questions as targets.

3.2.4 Procedure

Written instructions were provided prior to the actual task, followed by three practice trials which mimicked the exact procedure of the test trials. On each trial, participants were presented with a short written scenario, as described above. The context paragraph was presented automatically at the center of the screen together with the written cue "Press the SPACE bar to reveal the words". Each individual trial proceded as a standard moving window self-paced reading task (Just, Carpenter, & Woolley, 1982). Below the written cue there were underscores marking the location of the question and its preceding sentence, both of which were masked at the beginning of the trial. In order to reveal each word in the masked sentences, participants pressed the space bar, as indicated by the cue on screen.

Once participants read the last region in the target sentence, pressing the space key would trigger a button labeled "Next", which had to be clicked in order to advance to the next trial. Once every third trial, instead of the "Next" button, participants were presented with a comprehension question relating to the scenario they read. They were asked to choose between two response options and received written feedback regarding their choice, which had the sole purpose of keeping participants engaged with the task of reading for comprehension.

3.2.5 Region of interest

Recall that our working hypothesis was that the presence or absence of an epistemic bias would affect incremental question interpretation differently depending on the syntactic form of the target questions. Notice, however, that although the general prediction is that there should be differences between question types, and that these differences should emerge due to the processing of different negation forms, the account we're entertaining is underspecified with regard to the precise locus and time-course of such a differential effect on online interpretation. In other words, the theory, as it stands, does not generate precise predictions as to when in processing such a difference should emerge. Should any difference emerge as a direct effect of processing the question form itself ($_{\text{high}}$*Isn't ...*/ $_{\text{low}}$*Is she not ...*)? Or perhaps as an effect of processing, further downstream, the first semantic cue as to what the question is actually about ($_{\text{high}}$*Isn't she getting ...*/ $_{\text{low}}$*Is she not getting ...*)? Finally, recall that Tian et al. (2021) find that it is only by the final noun of their positive polar questions that listeners are making predictions about the underlying proposition.

Complicating matters even further, the difference in form between a high negation and a low negation in English means that the two structures are not directly comparable when it comes to the potential locus of the expected effect. Whereas the high negation form consists of a contraction between the auxiliary verb used to mark a canonical question in English and the negation marker usually employed to reverse the polarity of the utterance (*Isn't*), in the low negation form these two elements are not only disjoint but there is also a third element, namely a pronoun, intervening between the two (*Is she not*), which has implications for how the negated question form is processed linearly as the linguistic signal unfolds.

Because of the issues just raised, at this stage in the investigation of negated questions with biases it is an open, empirical question *whether* there is context-dependent facilita-

tion during online processing and, if so, *when* exactly during incremental interpretation such facilitation occurs, leaving open considerable researcher degrees of freedom concerning where in the linguistic signal to situate the effect of interest. For the analysis here reported, we set the regions of interest as spanning over the question form and up until the main verb ($_{\text{high}}$*Isn't she getting*/ $_{\text{low}}$*Is she not getting*). This diverges from the regions of interest originally defined in the study pre-registration[6], which spanned only the question forms themselves ($_{\text{high}}$*Isn't*/ $_{\text{low}}$*Is she not*) along with their potential spillover regions, which in the current analysis are targeted as critical regions of interest. This revision in the analysis plan is motivated by two independent reasons. First, in line with the pre-registered plan, the main verb serves as the spillover region for the negation marker in the low negation, meaning that an effect caused by processing *not* might only be observed one region downstream from it, at the main verb. Then, as per the revised analysis, defined *after* originally inspecting the data, it seems reasonable to assume that the effect of processing the negated question form, both in the high and low negation, might only become apparent further downstream in the sentence when interpreting the first meaningful lexical cue as to what the actual content of the question is, or in other words, only as comprehenders process the first semantic cue as to what state of affairs the question is targeting (e.g. whether or not someone is getting a cat). Given this picture, in what follows we are primarily interested in looking at the processing signatures measured up until the main verb, the predictions being as stated at the beginning of this section.

3.2.6 Results

The data and analysis scripts are available for inspection under `https://osf.io/7a2jk/`.

[6] `https://osf.io/3zk6d`, see analysis section (C1) as well as the analysis script.

3.2.7 Data cleaning

Prior to any analysis, the data set was cleaned according to criteria defined in the pre-registered analysis plan, which include both participant-based and response-based exclusion criteria. We first excluded the data from any participant who self-reported being a monolingual speaker of a language other than English (n=5). We then further excluded any participant who did not reach an accuracy of 80% on the comprehension questions (n=2). Finally, we excluded any participant who deviated in more than 30% of their trials in more than 2.5 standard deviations from the grand mean for a given condition, as measured at any individual sentence region (n=1). Altogether, these exclusions resulted in a data set which included data from 112 participants.

In addition to excluding seven individual participants as per the criteria above, we excluded any individual data point which deviated in more than 2.5 standard deviations from the grand mean for a given condition (0.01% of the data). We also removed any data point lying outside of a pre-established range of interest set between 100 and 1000 ms (0.02% of the data). This reflects a meaningful range for the reading of individual words in self-paced reading.

3.2.8 Confirmatory analyses

Figure 2 shows the main results, which are based on aggregates over all experimental items and participants. Crucially, recall that, by design, low negation items have one extra measurement region compared to high negation items, given the differences in syntactic form between both question types ($_{\text{high}}\underset{1}{Didn't}\ \underset{2}{you}\ \underset{3}{buy}/\ _{\text{low}}\underset{1}{Did}\ \underset{2}{you}\ \underset{3}{not}\ \underset{4}{buy}$).

In order to amass quantitative evidence in favor of the results reported below, we fitted Bayesian hierarchical models predicting RTs at the different critical regions as a function of a nested effect of belief scenario within negation type. This allows directly testing the effect of condition for each negation type, as per the paragraph below. The

models included the maximal random effect structure justified by the design, which in the present case is random intercepts for items – which varied systematically according to the belief scenario and negation type – and random slopes and intercepts for participants. These models, fitted using the R package `brms` (Bürkner, 2017) and described in detail in the scripts available in the supporting material, had the following general form, shown in `brms` syntax:

$$\begin{aligned}\log(\mathrm{RT}) \sim\ & \mathrm{condition} \ / \ \mathrm{negation} + \\ & (1 + \mathrm{condition} \ / \ \mathrm{negation} \mid \mathrm{participant}) + \\ & (1 \mid \mathrm{item})\end{aligned}$$

For each highlighted result, we report whether or not the respective statistical model provides strong evidence in favor of the empirically attested differences (or lack thereof). In a Bayesian statistical framework, one is interested in the joint posterior distribution of the parameters of the model, which indicates a plausible range of values for the parameters given the model and the data at hand. We report a 95% credibility interval (CrI) and the posterior probability that the parameter of interest, β, is smaller than zero ($P(\beta < 0)$). One speaks of strong evidence for an effect when zero is not included in the CrI and $P(\beta < 0)$ is close to either zero or one. Concretely, we are interested in the difference between estimated values for cell means of the two conditions, *Prior belief* and *No belief*, for each negation type. The theory here entertained predicts that, in the posterior distribution of the Bayesian regression model, the difference in cell means β_{high} = [estimates for cell mean of *Prior belief*] - [estimates for cell mean of *No belief*] should be credibly smaller than zero, so that $\mathrm{P}(\beta_{\text{high}} < 0)$ should be large, i.e., very close to 1; the opposite pattern is expected in the case of the low negation, that is, the difference in cell means β_{low} = [estimates for cell mean of *Prior belief*] - [estimates for cell mean of *No belief*] should be credibly bigger than zero, so that $P(\beta_{\text{low}} < 0)$ should be small, i.e., very close to 0.

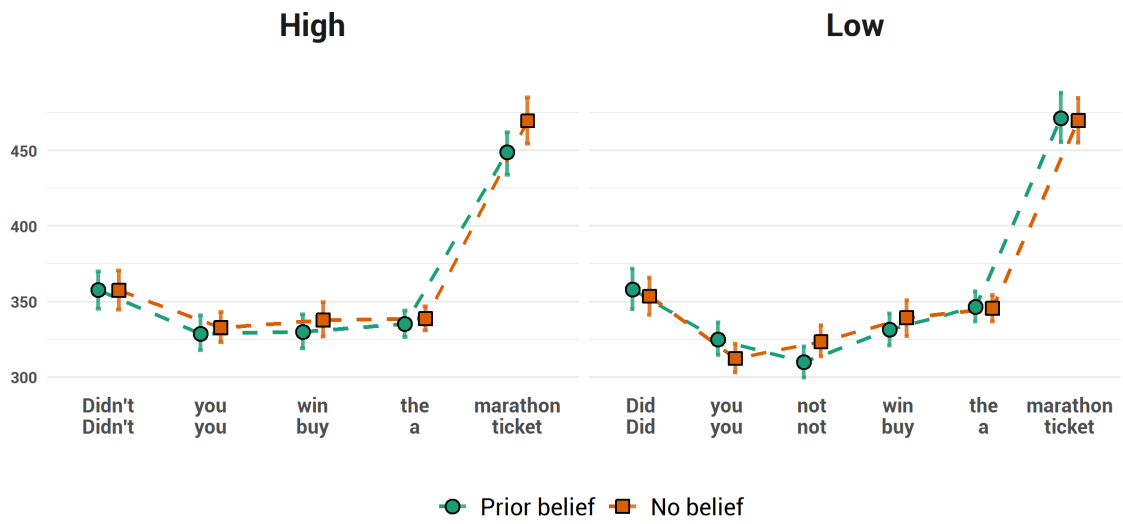

Figure 2: Mean reading times. Each column shows the grand mean for each each context type at each sentence region.

Visual inspection of the results suggests that, in the high negation, there are no systematic differences between the relevant *Prior belief* / *No belief* pairs at the two sentence regions preceding the main verb (see regression coefficients in Table 2). The same picture emerges at the main verb itself. In the low negation, much like in the high negation, there is no systematic difference between the relevant *Prior belief* / *No belief* pair at the auxiliary. At the pronoun, which follows the auxiliary and precedes the negation marker, the descriptive results suggest some facilitation for *No belief* items, as would be predicted for the low negation, however, this result finds no quantitative support in the statistical model ($\beta_{\text{low}} = 0.01$, 95% CrI [-0.02, 0.05], $P(\beta_{\text{low}} < 0) = 0.25$). Similarly, there seems to be some facilitation for *Prior belief* items at the negation marker, which would run counter to the prediction for the low negation, however, this result finds no quantitative support in the respective model. Finally, at the main verb, there is no systematic difference between the relevant *Prior belief* / *No belief* pair.

However, we do find one result with clear quantitative support: in the case of high negation questions, the last region (noun) in the sentence is read more slowly in *No belief*

scenarios compared to *Prior belief* ones (β_{high} = -0.07, 95% CrI [-0.12, -0.03], $P(\beta_{\text{high}} < 0) = 0.99$). This result gives some support to the hypothesis that high negation questions are facilitated when there is a prior belief associated with the underlying proposition (e.g. that someone won the marathon, or bought a ticket). The fact that it happens in the final region aligns with the findings of Tian et al. (2021), with facilitation for proposition processing happening late in the time course of the question.

Table 2: Regression coefficients.

Region	Hypothesis	β	95%-CrI	P($\beta < 0$)
Auxiliary	[high] Prior belief < No belief	0.00	[-0.03; 0.04]	0.42
Pronoun	[high] Prior belief < No belief	-0.03	[-0.06; 0.00]	0.94
Main verb	[high] Prior belief < No belief	-0.04	[-0.08; 0.00]	0.93
Verb spillover	[high] Prior belief < No belief	-0.03	[-0.07; 0.00]	0.96
Auxiliary	[low] Prior belief < No belief	0.01	[-0.03; 0.05]	0.32
Pronoun	[low] Prior belief < No belief	0.01	[-0.02; 0.05]	0.25
Negation marker	[low] Prior belief < No belief	-0.02	[-0.06; 0.02]	0.79
Main verb	[low] Prior belief < No belief	-0.02	[-0.05; 0.01]	0.84

3.3 Discussion

In Experiment 1, we set out to investigate the incremental interpretation of questions with biases in English. We addressed this issue using a task in which participants read negated polar questions against discourse contexts which contained different epistemic biases relating to the propositions targeted in the questions. Biases were operationalized as a combination of contextual evidence against the relevant proposition, on the one hand, and the presence or absence of a particular belief pertaining to the proposition, on the other. Our predictions were that these biases would interact with the form of a question – more specifically, the syntactic form of the negated questions – to give rise to differential effects on the measured reading times.

More concretely, we expected reading times at the critical sentence regions – which spanned from the auxiliary verb (*Didn't/ Did*) to the main verb (Didn't/ Did you not

buy) – to be lower when a high negation question was read in a discourse context containing a particular belief, and similarly so when a low negation question was read in a discourse context containing no particular belief. The hypothesized link between the empirical reading signatures and the underlying processing mechanism states that a particular word is harder to process – and hence takes longer to read – the less predictable that word is in context, in our case predictability being, by design, assumed to be tied to explicit expectations as per the constraints described above.

The quantitative analyses reported in the previous sections do not support these predictions. Despite not supporting the predictions with respect to the regions of interest, the results show evidence of facilitatory processing at the last sentence region in high negation questions read against contexts containing a questioner belief. We interpret this empirical observation as evidence that comprehenders might be able to track epistemic information when processing negated polar questions in English, despite not being sensitive to how that information might interact with different negation forms and the pragmatic biases arguably associated with them, at least not as evidenced in the current data.

4. Experiment 2 (German)

Experiment 2 follows the design and procedure of Experiment 1, focusing on the processing of biased questions in German. The reported study was pre-registered under `https://osf.io/hk3pb` and, unless otherwise noted, all analytic choices and procedures follow the pre-registered analysis plan.

4.1 Item norms

The materials consisted of translated and modified versions of the English items used in Experiment 1. Given potential effects of translation and adaptation, we again normed

the discourse contexts used in the processing task to make sure that they gave rise to the epistemic expectations we intended. 40 self-reported native speakers of German were recruited via the crowdsourcing platform Prolific (the data from one participant failed to be submitted to the database). The norming task followed the same procedure as detailed for English in section 3.1.

Figure 3 shows the item norms for the German materials aggregated over all items and participants. Participants interpreted *Prior belief* items, on average, as reflecting specific beliefs about the discourse scenarios in those same items. By the same token, they interpreted *No belief* items, on average, as reflecting no particular prior beliefs about the discourse scenarios in those items. All in all, these results show that the items do reflect the general epistemic expectations which are of theoretical relevance to the account of question interpretation entertained in this study.

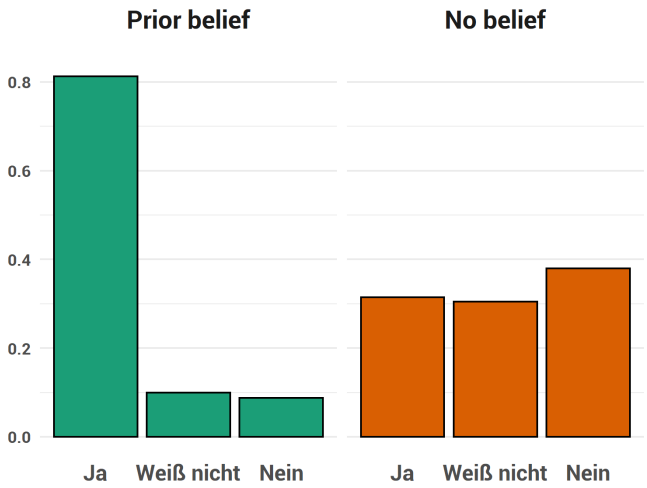

Figure 3: Item norms plotted by context type, *Prior belief* context shown in green, *No belief* context shown in orange. Each bar shows the proportion of choices of each response option, labeled individually below the respective bars.

4.2 Processing task

4.2.1 Method

4.2.2 Participants

80 native speakers of German were recruited among the cognitive science student population of the Osnabrück University. Participants were given course credit for their participation in the experiment. Data collection was conducted entirely online and all collected data was stored at servers from the Osnabrück University.

4.2.3 Materials, design, and procedure

The design and procedure were the same as in the processing task used in Experiment 1. The materials consisted of translated and modified versions of the English items used in Experiment 1, as exemplified in (12) and (13) below.

(12) **Example item (*Prior belief, high negation*)**

Marie, eine Freundin von uns, will ein neues Haustier adoptieren. Ihre Schwester hat dir erzählt, es wird eine Katze. Ich erzähle dir allerdings, dass sie vorhat eine Menge Spaziergänge mit dem Haustier zu unternehmen. Du sagst:
Warte mal. Wollte Marie nicht eine Katze haben?

Marie, a friend of ours, wants to adopt a cat. Her sister told you that it will be a cat. I tell you however that she's planning to take her pet on loads of of walks. You say:
Wait a second. Didn't Marie want a cat?

In (12), *Ihre Schwester hat dir erzählt, es wird eine Katze* is the belief, *Ich erzähle dir*

allerdings, dass sie vorhat eine Menge Spaziergänge mit dem Haustier zu unternehmen is the negative contextual evidence, and *Wollte Marie nicht eine Katze haben?* is the target sentence. Each target sentence conformed to the sentence frame *auxiliary* | *pronoun/ proper name* | *(negation marker)* | *(negated) article* | *noun* | *noun spillover* | *verb*, and was preceded by a short declarative in order to ensure that the first region was not a region of interest. In (13) below, *Wir unterhalten uns über ein Programm in unserem Lieblingstheater* is the utterance without any belief, *Ich sage, dass ich nicht hingehen werde* is the negative contextual evidence, and *Hast du nicht ein Ticket gekauft?* is the target sentence.

(13) **Example item (*No belief, high negation*)**

Wir unterhalten uns über ein Programm in unserem Lieblingstheater. Ich sage, dass ich nicht hingehen werde. Du sagst:
Wie schade. Hast du nicht ein Ticket gekauft?

We are talking about a performance at our favorite theater. I say that I'm not going. You say:
What a pity. Didn't you buy a ticket?

4.2.4 Hypotheses

Our working hypothesis was the same as in Experiment 1, namely that the presence or absence of an epistemic bias would affect incremental question interpretation differently depending on the syntactic form of the target questions. Notice that, much like in English, the two question forms in German do not have the same number of words. Whereas the high negation form consists of a sentential negation where the negation marker is placed before an indefinite article (*Hast du nicht ein*), the low negation form consists of a morphologically negated indefinite article (*Hast du kein*).

Just like in Experiment 1, then, the regions of interest span over the question form and up until the first semantic cue in the sentence. In German, the first semantic cue is the noun following the (potentially negated) determiner, with the main verb coming in final position($_{\text{high}}$*Hast du nicht ein Ticket*/ $_{\text{low}}$*Hast du kein Ticket*). The predictions are as stated at the beginning of this section.

4.2.5 Results

The data and analysis scripts are available for inspection under `https://osf.io/2ndf3/`.

4.2.6 Data cleaning

Prior to any analysis, the data set was cleaned according to criteria defined in the pre-registered analysis plan, as in Experiment 1. We first excluded the data from any participant who did not reach an accuracy of 80% on the comprehension questions (n=2). Finally, we excluded any participant who deviated in more than 30% of their trials in more than 2.5 standard deviations from the grand mean for a given condition, as measured at any individual sentence region (n=1). Altogether, these exclusions resulted in a data set which included data from 77 participants.

In addition to excluding three individual participants as per the criteria above, we excluded any individual data point which deviated in more than 2.5 standard deviations from the grand mean for a given condition (0.02% of the data). We also removed any data point lying outside of a pre-established range of interest set between 100 and 1000 ms (0.01% of the data). This reflects a meaningful range for the reading of individual words in self-paced reading.

4.2.7 Confirmatory analyses

Figure 4 shows the main results, which are based on aggregates over all experimental items and participants. Crucially, recall that, by design, high negation items have one extra measurement region compared to low negation items, given the differences in syntactic form between both question types ($_{\text{high}}\underset{1}{Hast}\ \underset{2}{du}\ \underset{3}{nicht}\ \underset{4}{ein}/\ _{\text{low}}\underset{1}{Hast}\ \underset{2}{du}\ \underset{3}{kein}$). In order to amass quantitative evidence in favor of the results reported below, we fitted Bayesian hierarchical models predicting RTs at the different critical regions as a function of a nested effect of belief scenario within negation type. This allows directly testing the effect of condition for each negation type, as per the paragraph below. The models included random intercepts for items – which varied systematically according to the belief scenario and negation type – and random slopes and intercepts for participants. These models, fitted using the R package `brms` (Bürkner, 2017) and described in detail in the scripts available in the supporting material, had the following general form, shown in `brms` syntax:

$$\begin{aligned}\log(\text{RT}) \sim\ & \text{condition / negation} + \\ & (1 + \text{condition / negation} \mid \text{participant}) + \\ & (1 \mid \text{item})\end{aligned}$$

For each highlighted result, we report whether or not the respective statistical model provides strong evidence in favor of the empirically attested differences (or lack thereof). In a Bayesian statistical framework, one is interested in the joint posterior distribution of the parameters of the model, which indicates a plausible range of values for the parameters given the model and the data at hand. We report a 95% credibility interval (CrI) and the posterior probability that the parameter of interest, β, is smaller than zero ($P(\beta < 0)$). One speaks of strong evidence for an effect when zero is not included in the CrI and $P(\beta < 0)$ is close to either zero or one. Concretely, we are interested in

the difference between estimated values for cell means of the two conditions, *Prior belief* and *No belief*, for each negation type. The theory here entertained predicts that, in the posterior distribution of the Bayesian regression model, the difference in cell means β_{high} = [estimates for cell mean of *Prior belief*] - [estimates for cell mean of *No belief*] should be credibly smaller than zero, so that $P(\beta_{\text{high}} < 0)$ should be large, i.e., very close to 1; the opposite pattern is expected in the case of the low negation, that is, the difference in cell means β_{low} = [estimates for cell mean of *Prior belief*] - [estimates for cell mean of *No belief*] should be credibly bigger than zero, so that $P(\beta_{\text{low}} < 0)$ should be small, i.e., very close to 0.

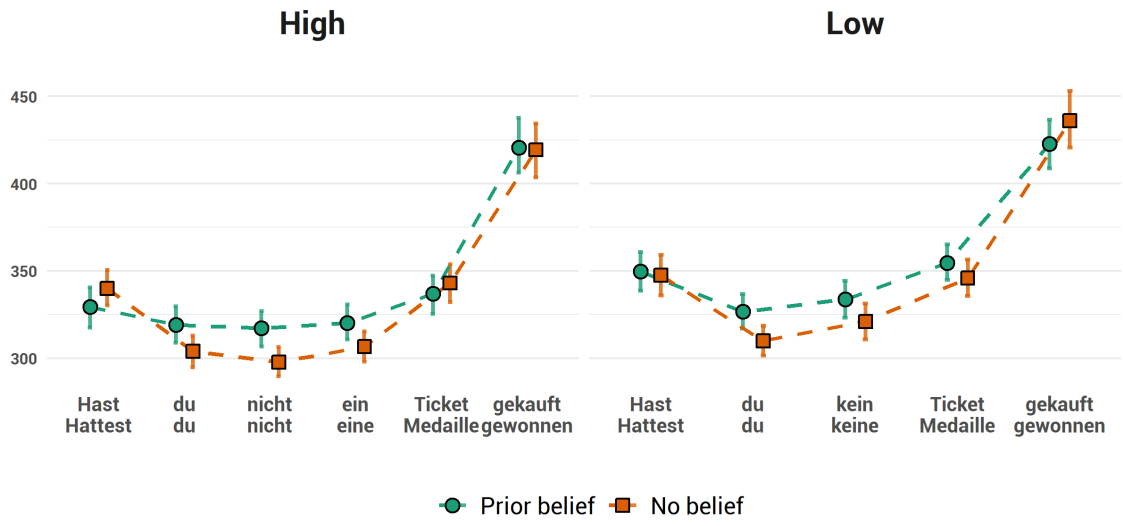

Figure 4: Mean reading times. Each column shows the grand mean for each each context type at each sentence region.

Visual inspection of the results suggests that, in the high negation, *Prior belief* items are read faster than *No belief* items at the auxiliary verb. This result, however, finds only weak quantitative support in the respective statistical model (β_{high} = -0.03, 95% CrI [-0.08, 0.01], $P(\beta_{\text{high}} < 0) = 0.87$). At the three regions following the auxiliary and preceding the noun, the results show that *No belief* items are read faster than *Prior belief* items, contrary to the prediction for the high negation. All these results find

strong quantitative support in the respective statistical models (see regression coefficients in Table 3). At the noun, the descriptive result suggests that there is no systematic difference between the *Prior belief* / *No belief* pair. In the low negation, visual inspection of the results suggests that there is no systematic difference between the relevant *Prior belief* / *No belief* pair at the auxiliary. At the two regions following the auxiliary and preceding the noun, the results show that *No belief* items are read faster than *Prior belief* items, in line with the prediction for the low negation (see regression coefficients in Table 3). Notice, however, that the result at the article finds weaker quantitative support compared to the result at the pronoun (see regression coefficients in Table 3 below). At the noun, while the descriptive result might suggest that *No belief* items are read faster than *Prior belief* items, in line with what was predicted, the result finds no quantitative support in the respective statistical model ($\beta_{\text{high}} = 0.01$, 95% CrI [-0.03, 0.06], $P(\beta_{\text{high}} < 0) = 0.32$).

While we did not have any predictions for the reading patterns at the main verb, the last region in the sentence, given the results of Experiment 1, one might expect that in German high negation questions the verb would also be read faster in *Prior belief* scenarios compared to *No belief* ones. This is, however, not the case, even though in the low negation the descriptive result seem to suggest an effect (which is not supported by the statistical model).

4.3 Discussion

In Experiment 2, we set out to investigate the incremental interpretation of questions with biases in German. Crucially, just like in Experiment 1, we addressed this issue using a task in which participants read negated polar questions against discourse contexts which contained different epistemic biases relating to the propositions targeted in the questions. Biases were operationalized as a combination of contextual evidence against

Table 3: Regression coefficients.

Region	Hypothesis	β	95%-CrI	P($\beta < 0$)
Auxiliary	[high] Prior belief < No belief	-0.03	[-0.08, 0.01]	0.87
Pronoun	[high] Prior belief < No belief	0.04	[0.00, 0.08]	0.03
Negation marker	[high] Prior belief < No belief	0.05	[0.01, 0.09]	0.03
Article	[high] Prior belief < No belief	0.05	[0.01, 0.08]	0.02
Noun	[high] Prior belief < No belief	-0.04	[-0.23, 0.15]	0.66
Auxiliary	[low] Prior belief < No belief	0.01	[-0.03, 0.06]	0.32
Pronoun	[low] Prior belief < No belief	0.05	[0.01, 0.09]	0.03
Negated article	[low] Prior belief < No belief	0.04	[0.00, 0.08]	0.07
Noun	[low] Prior belief < No belief	0.01	[-0.03, 0.06]	0.32
Main verb	[low] Prior belief < No belief	-0.01	[-0.08, 0.05]	0.63

the relevant proposition, on the one hand, and the presence or absence of a particular belief pertaining to the proposition, on the other. Our predictions were that these biases would interact with the syntactic form of the negated questions to give rise to differential effects on the measured reading times.

More concretely, we expected reading times at the critical sentence regions – which spanned from the auxiliary verb (*Hast*) to the noun (Hast du nicht ein/ Hast du kein *Ticket*) – to be lower when a high negation question was read in a discourse context containing a particular belief, and similarly so when a low negation question was read in a discourse context containing no particular belief. The quantitative analyses reported in the previous section are mixed with regard to these predictions. In the high negation there was primarily evidence against the prediction, with only weak evidence for the expected effect at the auxiliary, evidence against the expected effect in the critical regions between the auxiliary and the noun, and no reliable difference at the noun. In the low negation, on the other hand, there was no reliable difference at the auxiliary, evidence for the expected effect in the regions between the auxiliary and the noun, and no reliable difference at the noun. Moreover, unlike in Experiment 1, there was no evidence of facilitatory processing in high negation questions at the last sentence region.

All in all, much like in English, we interpret these results as evidence that comprehen-

ders might potentially track biases to do with the epistemic state of a questioner when processing negated polar questions in German. One observation which may reconcile the positive results attested in the low negation with the negative results attested in the high negation has to do with the fact that the tense of the auxiliary construction was often different between *Prior belief* and *No belief* items, as seen in the example items shown on the x-axis of Figure 4. While this may have confounded any potential effects of belief type, the decision to design the items using different verb tenses was made to ensure that they sounded as pragmatically appropriate as possible given the constraints of the task. It may very well be that the tense of the auxiliary served as an additional, if not conflicting, cue with regard to the relevant questioner biases. A similar result was found in Di Maro et al. (2021): for their Italian participants, there was a preference to produce high negation questions using the past tense, as opposed to the present. These results in general raise questions about the exact pragmatic considerations underlying the empirically attested production preferences, and how those might relate to particular linguistic forms.

5. General discussion

5.1 Implications for psycholinguistic accounts of online (question) processing

With regard to what the combined results of Experiments 1 and 2 might mean in terms of online processing, given the very limited evidence in the data for bias-related pragmatic prediction, it seems reasonable to conclude that in a task such as the present one, comprehenders do not track any biases to do with the epistemic state of a questioner — at least, not as they might compared to more naturalistic/interactive scenarios of language comprehension. Indeed, while other cues derived from the syntactic form of an

utterance have been shown to affect online sentence processing in the lab (e.g. Yano and Koizumi (2018)), the biases entertained here do not relate to straightforward referential tracking but rather to more social aspects of language interpretation, such as how a comprehender might *respond* to a question which is biased towards a particular state of affairs, or how an original speaker bias, as discussed in the literature, might affect what a comprehender *perceives* the epistemic/evidential support to a particular state of affairs to actually be. While response formation was not operationalized in the current study, the findings of the processing tasks seem at odds with the data from the norming tasks, which suggests that the discourse contexts built into the materials do reflect the theoretically-expected biases of interest. However, one should keep in mind that people's offline interpretation preferences, as reflected in the norming data, might not necessarily map onto expectations which are updated incrementally during processing, at least not at the granularity which can be captured using a reaction-time-based method like word-by-word self-paced reading.

Moreover, while negated polar questions do encode a questioner's stance in relation to the expected likelihood of a given response, both in the lab and in the wild these epistemic biases might be secondary in the face of a question's primary pragmatic function – which might be to request information, invite someone somewhere, or request a favor (Heritage and Raymond (2021)). And while clear-cut behavioral data with respect to the production of such questions has been shown to be elicitable in the lab (Di Maro et al. (2021); Domaneschi et al. (2017)), in online processing the relevant pragmatic biases might only become apparent in light of stronger, potentially more ecologically valid comprehension pressures. Indeed, one seemingly robust finding of recent studies on (probabilistic) predictive processing (e.g. Roettger and Franke (2019)), is that rational processing demands and their accompanying behavioral correlates might not necessarily map onto categorical (Gricean) pragmatic expectations unless there are strong enough

pressures in place such that the relevant expectations, which seem to be probabilistic in nature, actually become resource-rational in context.

All in all, the question of whether epistemic biases can impact incremental question interpretation should be further investigated in the lab. A particularly fruitful avenue for future research relates exactly to the conditions under which tracking biases and predicting upcoming material in an utterance might become pragmatically relevant. In the current study, the experimental task involved reading questions for comprehension, thus not involving any sort of actual contingent response preparation or decision-making. In other words, reading for comprehension might not give rise to enough of a pressure such that comprehenders track potential biases encoded by the form of a question, even if these biases might, in principle, be pragmatically relevant.

Given that polar questions seem particularly well-suited to investigations in terms of (dis)preferred responses, future experimental studies could situate question interpretation not only in more or less felicitous discourse contexts but, crucially too, in pragmatically and/or otherwise socially accountable action scenarios (see Benz and Gotzner (2020); Gisladottir, Chwilla, and Levinson (2015) for examples of such paradigms in experimental pragmatic research). In fact, the empirical observation that the participants in Experiment 1 tracked the beliefs built into the discourse situations when processing high negation questions is evidence that comprehenders are sensitive to pragmatic biases in question interpretation, only that those biases might show in a gradient manner, depending on the relevant pressures at play during comprehension.

5.2 Implications for theoretical accounts of question interpretation

While the current results are at best inconclusive with regard to comprehenders' online sensitives to the interpretive biases associated with the form of a negated polar question,

they do shed some light onto the question of what type of interpretation process might be at play in the processing of questions with biases, and what meaning might be more strongly associated with them.

Recall in section 2, we saw that the morphosyntacic form of the negated question in both English and German—specifically, whether the negation was *high* or *low* in the question—was claimed to encode either *outer* or *inner* negation, with the licensing of the question reliant on a particular combination of *original speaker bias* and *contextual evidence*. Low negation (*not* or *kein(e)*) was claimed to be licensed only when the speaker had no original bias towards the truth of a proposition p, but was presented with some contextual evidence suggesting that $\neg p$ was true, leading to an inner negation interpretation. High negation (*-n't* or *nicht ein(e)*) was claimed to be licensed when the speaker had an original bias that the proposition p was true, and, optionally, there may have been some negative contexual evidence that $\neg p$ was true—leading to a possible ambiguity between inner and outer negation. The claims made in the literature, as well as the results from production studies, indicated that an original speaker bias towards the proposition, or lack thereof, was essential to the production and, by extension, interpretation of the morphosyntax of a negated question.

On the basis of our data, interpreting actual epistemic biases seems not to be directly relevant to the processing of biased questions, at least not as relevant as parsing the utterance and tracking potential referents in discourse. Regarding the interpretation process potentially underlying the processing of questions with biases, it thus seems reasonable to assume that interpretation might involve some form of inferencing, as argued in, e.g. Van Rooy and Safarova (2003) (see also Lauer (2014)). It seems that any expectations associated with such questions might require strong contextual demands in order for them to systematically impact online interpretation, and not that there is a specific semantics associated with the high negation position that encodes an existing belief,

regardless of contextual evidence. Ultimately, epistemic biases might play a non-crucial role in the interpretation of questions, both in terms of strict semantic understanding and more generally in terms of recognizing any stances or pragmatic meanings signaled by questions in discourse.

However, note that in the present study, we only focused on the presence or absence of an original speaker bias in combination with negative contextual evidence against the proposition. The design thus heavily focused on the role of the variation in the morphosyntactic form of the question only within the context in which semantically inner negation is expected. It may be that focusing on the role of contextual evidence—and thus the distinction between a semantically outer negation, seeking confirmation of an existing belief, and the inner negation reading—might make interpreting epistemic biases more relevant when processing the different forms of negated questions.

Future research should thus tackle the role of contextual evidence in the processing of biased questions, in combination with the epistemic biases and the exact semantics proposed—exploring the circumstances in which one might affect the other, and perhaps more importantly, the circumstances in which one cannot be disentangled from the other. Furthermore, as discussed in the previous section, it may be that reading for comprehension is not a fine-grained enough measure to see expectations updated incrementally in the processing of biased questions, and alternative methods could better tap in to the complex relationship between beliefs, biases, semantics and morphosyntax involved in the interpretation of non-canonical questions.

6. Conclusions

In this study, we set out to investigate the hypothesis that the morphosyntactic form of negated polar questions in English and German signals epistemic biases which affect incremental question interpretation. We found limited evidence in support of that

hypothesis in our self-paced reading data, and on the basis of that we concluded that interpreting epistemic biases seems not to be directly relevant to the processing of biased polar questions, at least not as relevant as parsing the actual utterance and tracking potential referents in discourse. Instead, the data seems to suggest that incremental interpretation in such cases might involve some form of inferencing which is modulated by stronger pragmatic constraints in context.

References

Bartels, C. (1999). *The Intonation of English Statements and Questions.* Garland Publishers.

Benz, A., & Gotzner, N. (2020). Embedded implicature: what can be left unsaid? *Linguistics and Philosophy*, 1–32.

Biezma, M., & Rawlins, K. (2017). Rhetorical questions: Severing questioning from asking. In D. Burgdorf, J. Collard, S. Maspong, & B. Stefánsdóttir (Eds.), *Proceedings of SALT 27.* University of Maryland: LSA.

Büring, D., & Gunlogson, C. (2000). Aren't positive and negative questions the same? *Unpublished manuscript, UCSC/UCLA. Available online at: http://semantics archive. net/Archive/mYwOGNhO/polar_ questions. pdf*.

Bürkner, P.-C. (2017). brms: An r package for bayesian multilevel models using stan. *Journal of statistical software*, *80*(1), 1–28.

Dayal, V. (2016). *Questions.* Oxford University Press.

De Ruiter, J. P. (2012). *Questions: Formal, functional and interactional perspectives* (Vol. 12). Cambridge University Press.

Di Maro, M., Origlia, A., & Cutugno, F. (2021). Polar express: Polar question forms expressing bias-evidence conflicts in italian.

Domaneschi, F., Romero, M., & Braun, B. (2017). Bias in polar questions: Evidence from english and german production experiments. *Glossa: a journal of general linguistics*, *2*(1).

Farkas, D. F., & Roelofsen, F. (2017). Division of labor in the interpretation of declaratives and interrogatives. *Journal of Semantics*, *34*(2), 237–289.

Gisladottir, R. S., Chwilla, D. J., & Levinson, S. C. (2015, 03). Conversation electrified: Erp correlates of speech act recognition in underspecified utterances. *PLOS ONE*,

10(3), 1-24. doi: 10.1371/journal.pone.0120068

Groenendijk, J. A. G., & Stokhof, M. J. B. (1984). *Studies on the semantics of questions and the pragmatics of answers* (Unpublished doctoral dissertation). Univ. Amsterdam.

Gyuris, B. (2016). New perspectives on bias in polar questions: A study of Hungarian -e. *International Review of Pragmatics*, *8*(1), 1–50.

Hamblin, C. L. (1973). Questions in Montague English. *Foundations of Langauge*, *10*, 42–53.

Han, C.-h. (2002). Interpreting interrogatives as rhetorical questions. *Lingua*, *112*(3), 201–229.

Hedberg, N., Sosa, J. M., & Görgülü, E. (2017). The meaning of intonation in yes-no questions in American English: A corpus study. *Corpus Linguistics and Linguistic Theory*, *13*(2), 1–48.

Heritage, J., & Raymond, C. (2021). Preference and polarity: Epistemic stance in question design. *Research on Language and Social Interaction*, *54*(1), 39–59.

Jamieson, E., & Macuch-Silva, V. (to appear). Psycholinguistic processing tasks and the study of question bias. In A. Benz, M. Krifka, T. Trinh, & K. Yatushiro (Eds.), *Biased Questions: Experimental Results and Theoretical Modelling.* Berlin: Language Science Press.

Just, M., Carpenter, P., & Woolley, J. (1982). Paradigms and processes in reading comprehension. *Journal of Experimental Psychology: General*, *111*(2), 228–238.

Kaiser, E., & Trueswell, J. C. (2004). The role of discourse context in the processing of a flexible word-order language. *Cognition*, *94*(2), 113–147.

Karttunen, L. (1977). Syntax and semantics of questions. *Linguistics and philosophy*, *1*(1), 3–44.

Krifka, M. (2017). Negated polarity questions as denegations of assertions. In *Con-*

trastiveness in information structure, alternatives and scalar implicatures (pp. 359–398). Springer.

Kurumada, C., Brown, M., Bibyk, S., Pontillo, D. F., & Tanenhaus, M. K. (2014). Is it or isn't it: Listeners make rapid use of prosody to infer speaker meanings. *Cognition*, *133*(2), 335–342.

Ladd, D. R. (1981). A first look at the semantics and pragmatics of negative questions and tag questions. In *Papers from the... regional meeting. chicago ling. soc. chicago, ill* (pp. 164–171).

Lauer, S. (2014). Mandatory implicatures in gricean pragmatics. In *Formal & experimental pragmatics workshop, essli 2014* (pp. 21–28).

Raymond, G. (2003). Grammar and social organization: Yes/no interrogatives and the structure of responding. *American sociological review*, 939–967.

Roelofsen, F., Venhuizen, N., & Weidman Sassoon, G. (2012). Positive and negative questions in discourse. In E. Csipak & H. Zeijlstra (Eds.), *Proceedings of Sinn und Bedeutung 17* (pp. 519–536). Göttingen: University of Göttingen.

Roettger, T. B., & Franke, M. (2019). Evidential strength of intonational cues and rational adaptation to (un-) reliable intonation. *Cognitive science*, *43*(7), e12745.

Romero, M. (2020). Form and function of negative, tag, and rhetorical questions. In *The oxford handbook of negation.*

Romero, M., & Han, C.-h. (2004). On negative yes/no questions. *Linguistics and philosophy*, *27*(5), 609–658.

Sadock, J. (1974). *Towards a linguistic theory of speech acts.* New York: Academic Press.

Sailor, C. (2013). *Questionable negation.* Presentation at LSA Annual Meeting.

Searle, J. R. (1969). *Speech acts: An essay in the philosophy of language* (Vol. 626). Cambridge university press.

Stenström, A.-B. (1984). *Questions and responses in english conversation.* Krieger Pub Co.

Sudo, Y. (2013). Biased polar questions in english and japanese. In *Beyond expressives: Explorations in use-conditional meaning* (pp. 275–295). Brill.

Thoms, G., Adger, D., Heycock, C., Jamieson, E., & Smith, J. (2023). English contracted negation revisited: Evidence from varieties of scots. *Language*, *99*(4), 726–759.

Tian, Y., van Tiel, B., Clin, E., & Breheny, R. (2021). Representing polar questions. *Journal of Psycholinguistic Research*, *50*, 1535–1555.

Van Rooy, R., & Safarova, M. (2003). On polar questions. In *Semantics and linguistic theory* (Vol. 13, pp. 292–309).

Yano, M., & Koizumi, M. (2018). Processing of non-canonical word orders in (in) felicitous contexts: Evidence from event-related brain potentials. *Language, Cognition and Neuroscience*, *33*(10), 1340–1354.